



CAT 2025 Slot 3 Question Paper - QA

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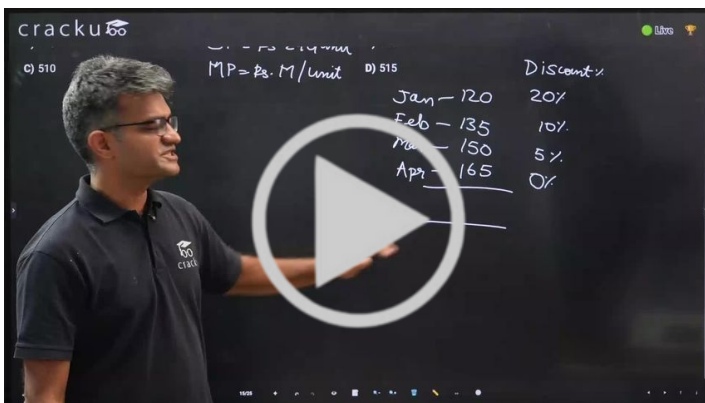
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Quant

1. The monthly sales of a product from January to April were 120, 135, 150 and 165 units, respectively. The cost price of the product was Rs. 240 per unit, and a fixed marked price was used for the product in all the four months. Discounts of 20%, 10% and 5% were given on the marked price per unit in January, February and March, respectively, while no discounts were given in April. If the total profit from January to April was Rs. 138825, then the marked price per unit, in rupees, was

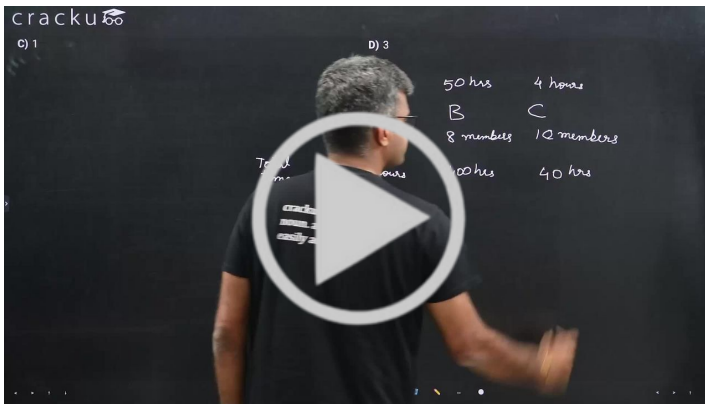
- A 520
- B 525
- C 510
- D 515



[VIDEO SOLUTION](#)

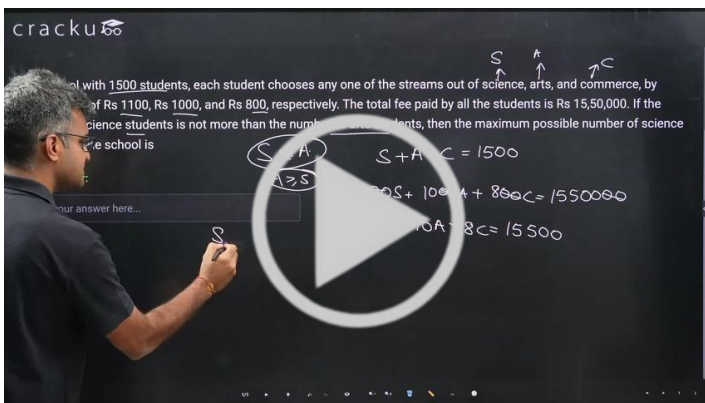
2. Teams A, B, and C consist of five, eight, and ten members, respectively, such that every member within a team is equally productive. Working separately, teams A, B, and C can complete a certain job in 40 hours, 50 hours, and 4 hours, respectively. Two members from team A, three members from team B, and one member from team C together start the job, and the member from team C leaves after 23 hours. The number of additional member(s) from team B, that would be required to replace the member from team C, to finish the job in the next one hour, is

- A 4
- B 2
- C 1
- D 3



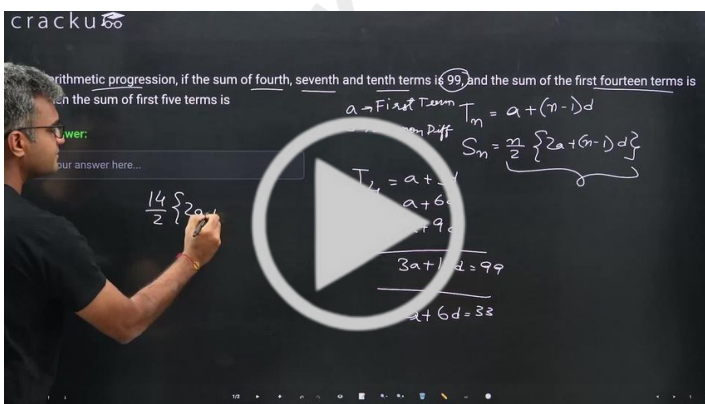
VIDEO SOLUTION

3. In a school with 1500 students, each student chooses any one of the streams out of science, arts, and commerce, by paying a fee of Rs 1100, Rs 1000, and Rs 800, respectively. The total fee paid by all the students is Rs 15,50,000. If the number of science students is not more than the number of arts students, then the maximum possible number of science students in the school is



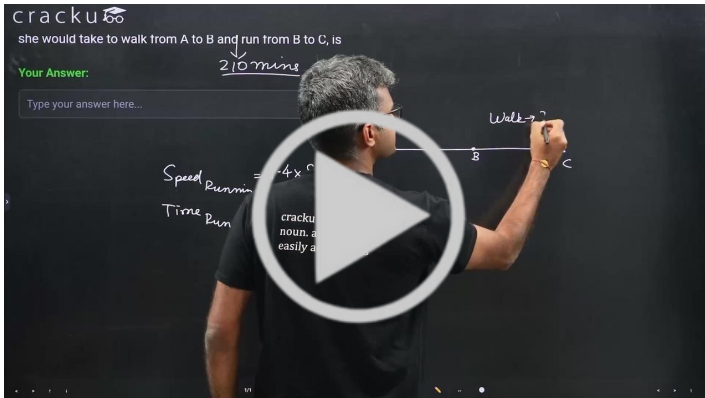
VIDEO SOLUTION

4. In an arithmetic progression, if the sum of fourth, seventh and tenth terms is 99, and the sum of the first fourteen terms is 497, then the sum of first five terms is



VIDEO SOLUTION

5. Ankita walks from A to C through B, and runs back through the same route at a speed that is 40% more than her walking speed. She takes exactly 3 hours 30 minutes to walk from B to C as well as to run from B to A. The total time, in minutes, she would take to walk from A to B and run from B to C, is



VIDEO SOLUTION

6. For a 4-digit number (greater than 1000), sum of the digits in the thousands, hundreds, and tens places is 15. Sum of the digits in the hundreds, tens, and units places is 16. Also, the digit in the tens place is 6 more than the digit in the units place. The difference between the largest and smallest possible value of the number is

- A 811
- B 3289
- C 735
- D 4078



VIDEO SOLUTION

7. Rahul starts on his journey at 5 pm at a constant speed so that he reaches his destination at 11 pm the same day. However, on his way, he stops for 20 minutes, and after that, increases his speed by 3 km per hour to reach on time. If he had stopped for 10 minutes more, he would have had to increase his speed by 5 km per hour to reach on time. His initial speed, in km per hour, was

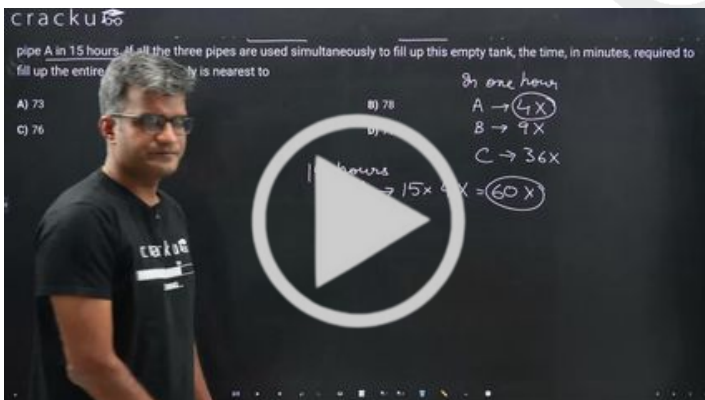
- A 12
- B 15
- C 18
- D 20



VIDEO SOLUTION

8. The rate of water flow through three pipes A, B and C are in the ratio 4 : 9 : 36. An empty tank can be filled up completely by pipe A in 15 hours. If all the three pipes are used simultaneously to fill up this empty tank, the time, in minutes, required to fill up the entire tank completely is nearest to

- A 73
- B 78
- C 76
- D 71



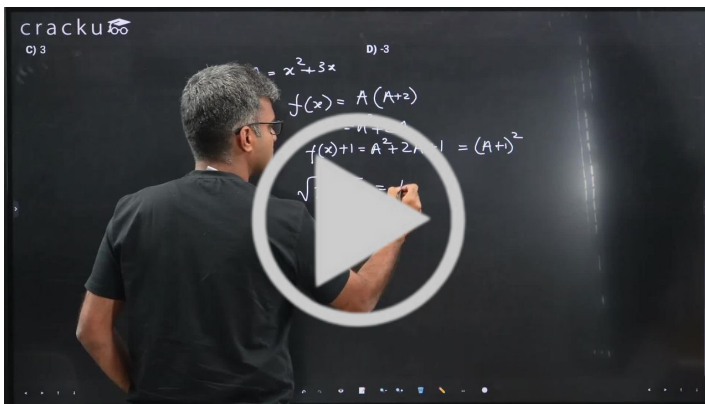
VIDEO SOLUTION

9. If $f(x) = (x^2 + 3x)(x^2 + 3x + 2)$ then the sum of all real roots of the equation $\sqrt{f(x) + 1} = 9701$, is

- A -6
- B 6

C 3

D -3



VIDEO SOLUTION

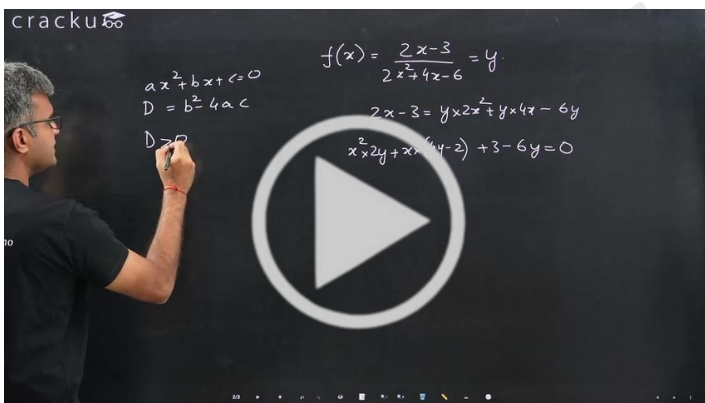
10. For real values of x , the range of the function $f(x) = \frac{2x-3}{2x^2+4x-6}$ is

A $(-\infty, \frac{1}{8}] \cup [1, \infty)$

B $(-\infty, \frac{1}{4}] \cup [1, \infty)$

C $(-\infty, \frac{1}{8}] \cup [\frac{1}{2}, \infty)$

D $(-\infty, \frac{1}{4}] \cup [\frac{1}{2}, \infty)$



VIDEO SOLUTION

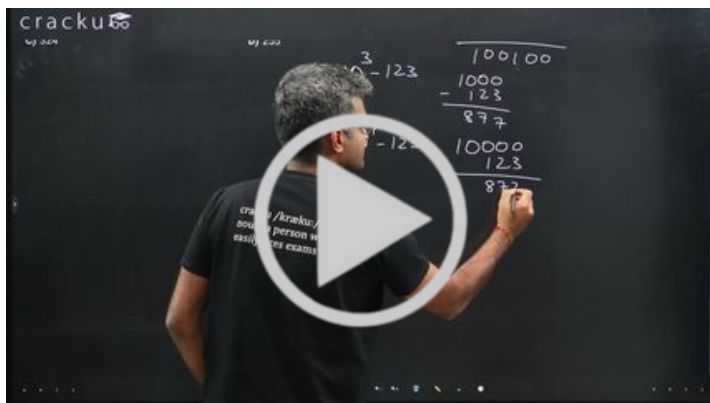
11. The sum of all the digits of the number $(10^{50} + 10^{25} - 123)$, is

A 212

B 221

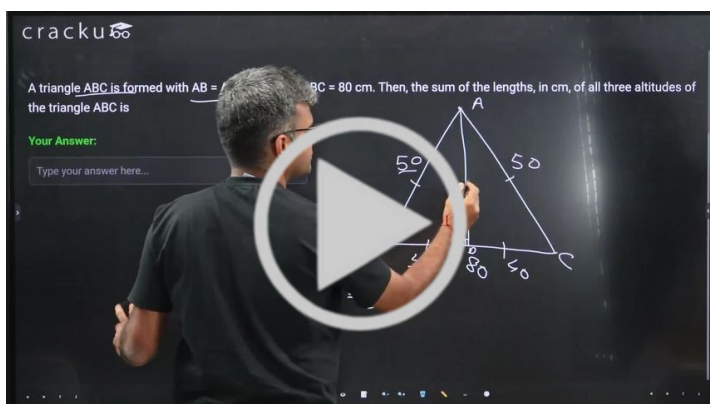
C 324

D 255



VIDEO SOLUTION

12. A triangle ABC is formed with $AB = AC = 50$ cm and $BC = 80$ cm. Then, the sum of the lengths, in cm, of all three altitudes of the triangle ABC is



VIDEO SOLUTION

13. If $(x^2 + \frac{1}{x^2}) = 25$ and $x > 0$, then the value of $(x^7 + \frac{1}{x^7})$ is

- A $44853\sqrt{3}$
- B $44856\sqrt{3}$
- C $44859\sqrt{3}$
- D $44850\sqrt{3}$



VIDEO SOLUTION

14. In a class of 150 students, 75 students chose physics, 111 students chose mathematics and 40 students chose chemistry. All students chose at least one of the three subjects and at least one student chose all three subjects. The number of students who chose both physics and chemistry is equal to the number of students who chose both chemistry and mathematics, and this is half the number of students who chose both physics and mathematics. The maximum possible number of students who chose physics but not mathematics, is

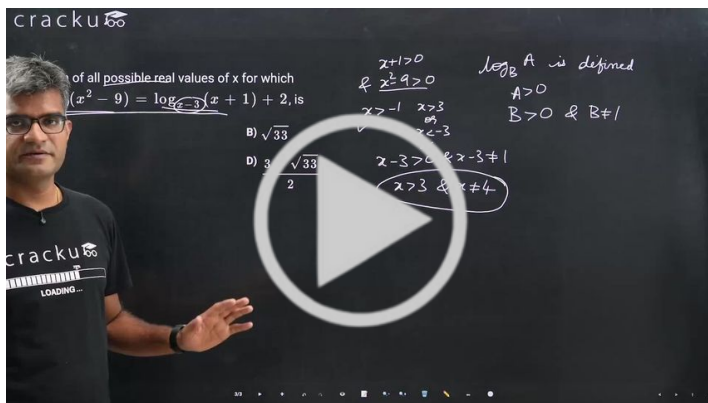
- A 30
- B 35
- C 40
- D 55



VIDEO SOLUTION

15. The sum of all possible real values of x for which $\log_{x-3}(x^2 - 9) = \log_{x-3}(x + 1) + 2$, is

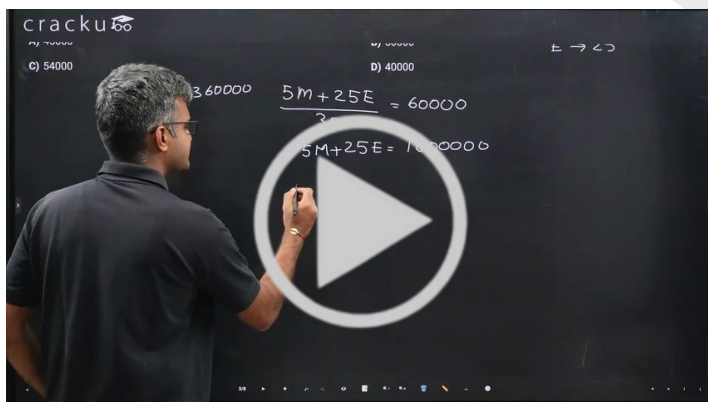
- A -3
- B $\sqrt{33}$
- C 3
- D $\frac{3 + \sqrt{33}}{2}$



VIDEO SOLUTION

16. The average salary of 5 managers and 25 engineers in a company is 60000 rupees. If each of the managers received 20% salary increase while the salary of the engineers remained unchanged, the average salary of all 30 employees would have increased by 5%. The average salary, in rupees, of the engineers is

- A 45000
- B 50000
- C 54000
- D 40000



VIDEO SOLUTION

17. ABCD is a trapezium in which AB is parallel to DC, AD is perpendicular to AB, and $AB = 3DC$. If a circle inscribed in the trapezium touching all the sides has a radius of 3 cm, then the area, in sq. cm, of the trapezium is

- A 48
- B $30\sqrt{3}$
- C $36\sqrt{2}$
- D 54



VIDEO SOLUTION

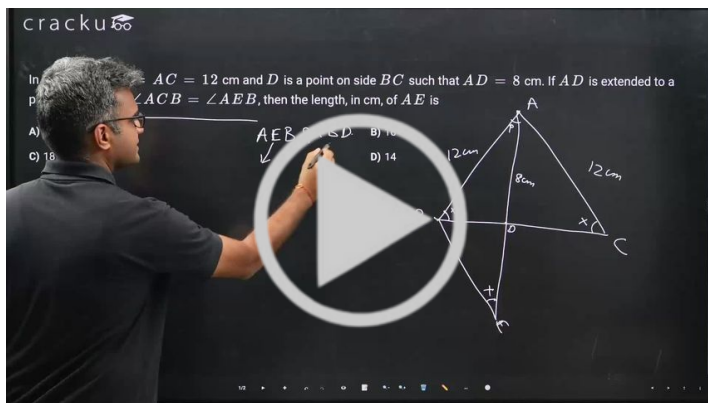
18. If $12^{12x} \times 4^{24x+12} \times 5^{2y} = 8^{4z} \times 20^{12x} \times 243^{3x-6}$, where x, y and z are natural numbers, then $x + y + z$ equals



VIDEO SOLUTION

19. In $\triangle ABC$, $AB = AC = 12$ cm and D is a point on side BC such that $AD = 8$ cm. If AD is extended to a point E such that $\angle ACB = \angle AEB$, then the length, in cm, of AE is

- A 20
- B 16
- C 18
- D 14



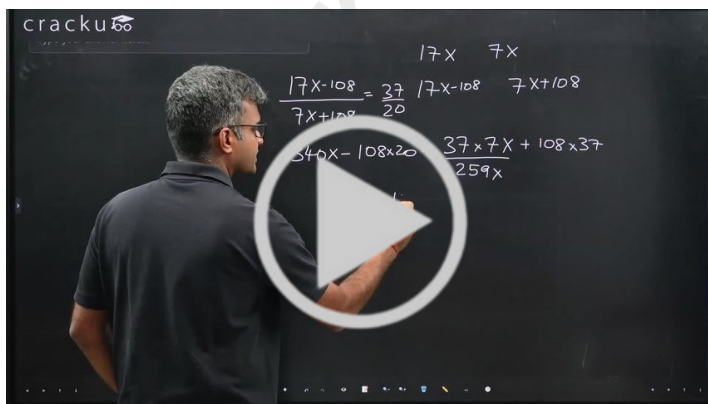
VIDEO SOLUTION

20. Vessels A and B contain 60 litres of alcohol and 60 litres of water, respectively. A certain volume is taken out from A and poured into B. After stirring, the same volume is taken out from B and poured into A. If the resultant ratio of alcohol and water in A is 15 : 4, then the volume, in litres, initially taken out from A is



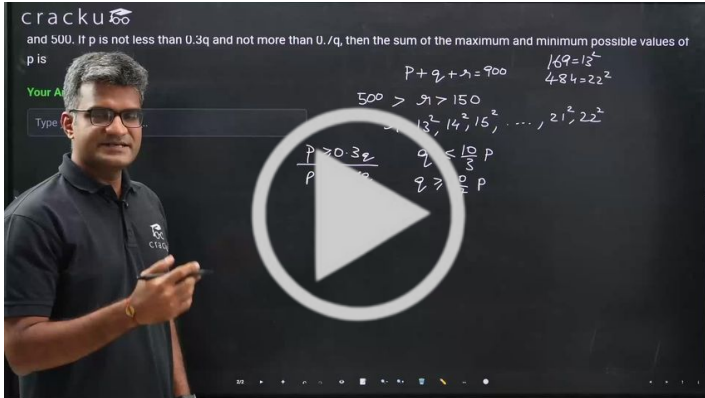
VIDEO SOLUTION

21. The ratio of the number of coins in boxes A and B was 17:7. After 108 coins were shifted from box A to box B, this ratio became 37:20. The number of coins that needs to be shifted further from A to B, to make this ratio 1:1, is



 VIDEO SOLUTION

22. Let p , q and r be three natural numbers such that their sum is 900, and r is a perfect square whose value lies between 150 and 500. If p is not less than $0.3q$ and not more than $0.7q$, then the sum of the maximum and minimum possible values of p is



 VIDEO SOLUTION

Answers

Quant

1.B	2.B	3.700	4.65	5.444	6.A	7.B	8.A
9.D	10.C	11.B	12.126	13.A	14.B	15.D	16.C
17.A	18.112	19.C	20.16	21.272	22.397		

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Explanations

Quant

1. B

The total number of products is $120 + 135 + 150 + 165 = 570$. Therefore, the total cost must have been $570 \times 240 = 136800$

Let the marked price for the products be X . The January, February, March, and April, selling prices would respectively be $0.8X$, $0.9X$, $0.95X$, and X ; based on the discounts given on each of the months.

The revenue earned from selling the products in each of the months would be: $120 * 0.8X + 135 * 0.9X + 150 * 0.95X + 165 * X = 525X$.

The profit earned would be: Revenue - Cost, and therefore,

$$525X - 136800 = 138825$$

$$525X = 275625$$

$$X = \frac{275625}{525} = 525$$

The correct answer is option B, Rs. 525.

 VIDEO SOLUTION

2. B

Let the per hour productivity of each member from teams A, B, and C, be a , b , and c units respectively.

Let W be the total units of work that each of the teams does. We have:

$$W = 40 * 5 * a = 50 * 8 * b = 4 * 10 * c$$

$$\text{This gives, } a = \frac{W}{200}, b = \frac{W}{400}, \text{ and } c = \frac{W}{40}$$

When two members from team A, three members from team B, and one member from team C together start the job, the total per hour productivity for the first 23 hours will be;

$$\frac{2W}{200} + \frac{3W}{400} + \frac{W}{40} = \frac{17W}{400}$$

$$\text{In 23 hours, the work done will be } \frac{23 * 17W}{400} = \frac{391W}{400}$$

$$\text{The remaining work, after 23 hours, is } W - \frac{391W}{400} = \frac{9W}{400}$$

$$\text{The current per hour efficiency of the group that comprises members from only teams A and B, is } \frac{2W}{200} + \frac{3W}{400} = \frac{7W}{400}$$

In the next hour, the members from A and B will finish $\frac{7W}{400}$ units of work and $\frac{9W}{400} - \frac{7W}{400} = \frac{2W}{400}$ units of work would remain.

This work has to be finished by newly added members from team B in one hour, therefore, the number of new

$$\text{members required from team B would be } \frac{2W}{400} \div \frac{W}{400} = 2$$

Option B is the correct answer.

3.700

Let the total number of students who chose science, arts, and commerce streams be S , A , and C , respectively.

We have,

$$S + A + C = 1500, \text{ such that, } C = 1500 - S - A \dots(1)$$

Also, $1100S + 1000A + 800C = 15,50,000$, which can be simplified by dividing by 100;

$$11S + 10A + 8C = 15500 \dots(2)$$

Substituting the value of C from equation (1) in equation (2), we have

$$11S + 10A + 8(1500 - S - A) = 15500$$

$$\Rightarrow 3S + 2A = 3500$$

Since $S \leq A$, the maximum value of S (if possible), would occur when $S = A$, therefore,

$$5S = 3500, \text{ or } S = 700.$$

4.65

Let the first term of the arithmetic progression be a , and the common difference be d .

We have:

$$(a + 3d) + (a + 6d) + (a + 9d) = 99$$

$$\Rightarrow 3a + 18d = 99 \text{ or } a + 6d = 33 \dots(1)$$

We are also told that the sum of the first fourteen terms is 497,

$$\frac{14}{2}(2a + (14 - 1)d) = 497$$

$$\Rightarrow (2a + 13d) = \frac{497}{7}$$

$$\Rightarrow 2a + 13d = 71 \dots(2)$$

Solving equations (1) and (2), we get;

$$a = 3 \text{ and } d = 5$$

The first five terms would therefore be: 3, 8, 13, 18, 23, and their sum would be $3 + 8 + 13 + 18 + 23 = 65$.

5.444

Let the walking speed of Ankita be $5x$, implying that her running speed (which is 40% more than her walking speed), is $1.4 \times 5x = 7x$.

Thus, the ratio of her walking and running speeds is 5 : 7. Therefore, the ratio of the time taken by Ankita to cover a fixed distance walking and running would be 7 : 5.

She takes 3 hours 30 minutes, or 3.5 hours to walk from B to C. Since Ankita is running from B to C in the second scenario, her time will reduce inversely to the ratio of her speed. The time taken by her to run from B to

C would be $\frac{3.5}{7} \times 5 = 2.5$ hours.

She takes 3 hours 30 minutes, or 3.5 hours to run from A to B. Since Ankita is walking from A to B in the second scenario, her time will increase inversely to the ratio of her speed. The time taken by her to walk from A to B would be $\frac{3.5}{5} \times 7 = 4.9$ hours.

Therefore, the total time it takes for Ankita in the second scenario is $4.9 + 2.5 = 7.4$ hours. This gives $7.4 \times 60 = 444$ minutes.

 VIDEO SOLUTION

6. A

Let the thousands, hundreds, tens, and units digits of the number be a, b, c , and d , respectively.

The sum of the digits in the thousands, hundreds, and tens places is 15; $a + b + c = 15$ (1)

The sum of the digits in the hundreds, tens, and units places is 16; $b + c + d = 16$ (2)

Solving equations (1) and (2), we get $d = a + 1$.

The digit in the tens place is 6 more than the digit in the units place; $c = d + 6 = a + 1 + 6 = a + 7$ (3)

Substituting the value of c in terms of a in equation (1), we get; $a + b + a + 7 = 15$ or $b = 8 - 2a$

Thus, the number is: a $8 - 2a$ $a + 7$ $a + 1$

We know that since the number is a four digit number, $a \geq 1$. But all the four digits have to be less than or equal to 9 but greater than or equal to 0, thus,

$8 - 2a \geq 0$ gives $a \leq 4$ and $a + 7 \leq 9$ gives $a \leq 2$.

Thus, a can be 1 or 2.

The two possible values for the number, therefore, are, 1682 and 2493, when we put $a = 1$ and $a = 2$, respectively.

The difference between the only two possible values is, $2493 - 1682 = 811$

Option A is the correct answer.

 VIDEO SOLUTION

7. B

Let the initial speed of Rahul be x kilometres per hour. Since he travels the distance usually in 6 hours, (5 pm to 11 pm), the total distance should be $6 \times x = 6x$ kilometres.

Let the distance after which Rahul stops for some duration in both scenarios be y kilometres.

In the first scenario, he stops for 20 minutes or $\frac{1}{3}$ hours, therefore, the travel time would be $6 - \frac{1}{3} = \frac{17}{3}$ hours. We have,

$$\frac{k}{x} + \frac{6x - k}{x + 3} = \frac{17}{3}$$

$$\Rightarrow \frac{6x^2 + 3k}{x^2 + 3x} = \frac{17}{3}$$

$$\Rightarrow 18x^2 + 9k = 17x^2 + 51x$$

$$\Rightarrow x^2 = 51x - 9k \text{(1)}$$

Similarly, in the second scenario, he stops for $20 + 10 = 30$ minutes or $\frac{1}{2}$ hours, therefore, the travel time would be $6 - \frac{1}{2} = \frac{11}{2}$ hours. We have,

$$\frac{k}{x} + \frac{6x - k}{x + 5} = \frac{11}{2}$$

$$\Rightarrow \frac{6x^2 + 5k}{x^2 + 5x} = \frac{11}{2}$$

$$\Rightarrow 12x^2 + 10k = 11x^2 + 55x$$

$$\Rightarrow x^2 = 55x - 10k \dots(2)$$

From equations (1) and (2), we have;

$$55x - 10k = 51x - 9k \text{ or}$$

$$4x = k.$$

Substituting the value of k in equation (1), we have

$x^2 = 51x - 36x$ or, since x is positive, $x = 15$. Therefore, option B is the correct answer.

 VIDEO SOLUTION

8. A

Let the rate of flow of water from pipe A be $4x$ units per hour. From the ratio of the rates of flow of water (4:9:36), we have that the rate of flow of water from pipe B should be $9x$ units per hour, and from pipe C should be $36x$ units per hour.

Since pipe A fills the empty tank in 15 hours, the capacity of the tank must be $15 \times 4x = 60x$ units.

When all three pipes work together, their combined capacity would be $4x + 9x + 36x = 49x$ units per hour.

Therefore, the time it would take for the three pipes to fill the empty tank working together is: $\frac{60x}{49x}$ hours

In minutes, this time is equivalent to $\frac{60x}{49x} \times 60 \approx 73.47$ minutes.

Option A is the closest, and is the correct answer.

 VIDEO SOLUTION

9. D

Let $(x^2 + 3x)$ be equal to k . We have,

$$f(x) = k(k + 2) = k^2 + 2k$$

$$\text{Therefore, } \sqrt{f(x) + 1} = \sqrt{k^2 + 2k + 1} = \sqrt{(k + 1)^2} = k + 1 = 9701$$

We get $k = 9700$.

$$\text{Thus, } x^2 + 3x = 9700 \text{ or } x^2 + 3x - 9700 = 0$$

Since x is real, the discriminant of the above quadratic has to be greater than or equal to zero.

We find that $3^2 + 4 * 9700 \geq 0$ and therefore the quadratic has real roots.

The sum of the roots will be $-\frac{b}{a} = -\frac{3}{1} = -3$

Option D is the correct answer.

 VIDEO SOLUTION

10. C

$$\text{Let } y = \frac{2x - 3}{2x^2 + 4x - 6}$$

$$\Rightarrow 2yx^2 + 4yx - 6y = 2x - 3$$

$$\Rightarrow 2yx^2 + (4y - 2)x - 6y + 3 = 0$$

Equation (1) is a quadratic in x , where x is real. Therefore, the discriminant of the quadratic has to be greater than or equal to zero.

$$(4y - 2)^2 + 4 \times 2y \times (6y - 3) \geq 0$$

$$16y^2 + 4 - 16y - 24y + 48y^2 \geq 0$$

$$64y^2 - 40y + 4 \geq 0$$

$$16y^2 - 10y + 1 \geq 0$$

The roots of the quadratic above will be $\frac{10 \pm 6}{32} = \frac{1}{2}$ or $\frac{1}{8}$

Since the coefficient of y^2 is positive, the quadratic will be less than zero in the range $\left(\frac{1}{8}, \frac{1}{2}\right)$

The quadratic will be greater than or equal to zero otherwise.

Therefore, the domain the quadratic, or possible values of y , which is the range of $f(x)$, will be,

$$\left(-\infty, \frac{1}{8}\right] \cup \left[\frac{1}{2}, \infty\right)$$

Option C is the correct answer.

[VIDEO SOLUTION](#)

11. **B**

$(10^{50} + 10^{25} - 123)$ can be written as $10^{50} + (10^{25} - 123)$

$1000 - 123$ gives 877.

$10000 - 123$ gives 9877.

$100000 - 123$ gives 99877.

$1000000 - 123$ gives 999877.

And so on.

Considering $10^n - 123$, which has a total of n digits; in general, of the n total digits; the digit 7 is repeating 2 times, digit 8 is repeating 1 time, and digit 9 is repeating $(n - 3)$ times.

Also, when added to 10^{50} , the cumulative increase in the sum of the digits of the resultant number would be 1, which would come from the leftmost digit of the result. All other newly added digits will be zero.

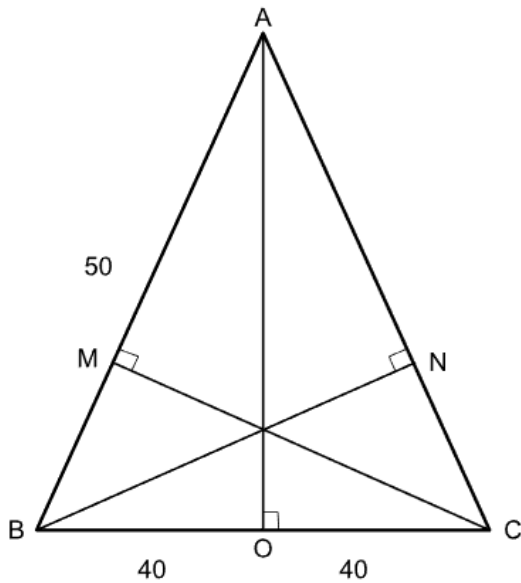
Thus, the sum of the digits of the result will be given by;

$$[7 * 2] + [1 * 8] + [(25 - 3) * 9] + [25 * 0] + [1 * 1] = 14 + 8 + 198 + 1 = 221$$

The correct answer is option B.

[VIDEO SOLUTION](#)

12. **126**



The altitudes CM and BN will be equal as the triangle $\triangle ABC$ is isosceles.

Further, in $\triangle AOB$, using Pythagoras Theorem, we have

$$AO^2 + OB^2 = AB^2, \text{ therefore, } AO^2 = 50^2 - 40^2$$

$$\text{Thus, } AO^2 = 900 \text{ or } AO = 30 \text{ cm.}$$

We know that the ratio of altitudes in a triangle is the inverse of the ratio of sides; therefore, the ratio of altitudes of the triangle will be;

$$\frac{1}{50} : \frac{1}{50} : \frac{1}{80} = 8 : 8 : 5$$

$$\text{Therefore, } CM = BN = 8x \text{ and } AO = 5x.$$

Since $5x = 30$, we have $x = 6$, and the sum of altitudes, in centimetres, will be $8x + 8x + 5x = 21x = 21 * 6 = 126$.

 VIDEO SOLUTION

13. A

$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = 25 + 2 = 27$$

$$\text{Therefore, } \left(x + \frac{1}{x}\right) = \sqrt{27} = 3\sqrt{3}$$

$$\text{Also, } \left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$$

$$\text{Therefore, } x^3 + \frac{1}{x^3} = (3\sqrt{3})^3 - 9\sqrt{3} = 72\sqrt{3}$$

$$\text{Also, } \left(x^2 + \frac{1}{x^2}\right)^2 = x^4 + \frac{1}{x^4} + 2$$

$$\text{Therefore, } x^4 + \frac{1}{x^4} = (25)^2 - 2 = 623$$

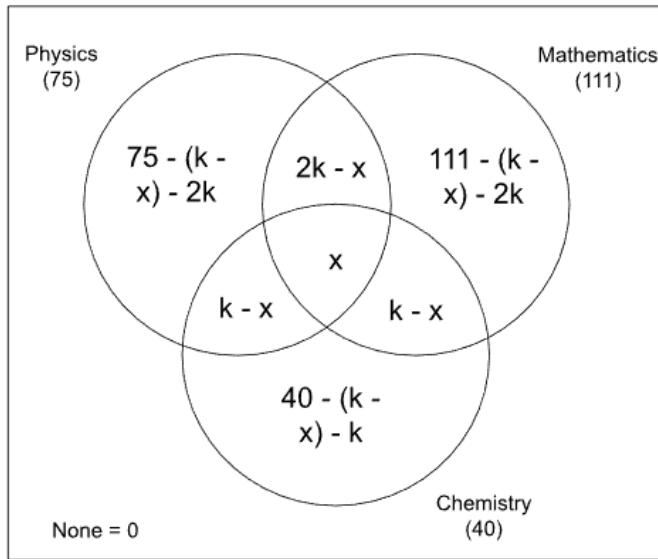
$$\text{Lastly, } \left(x^4 + \frac{1}{x^4}\right)\left(x^3 + \frac{1}{x^3}\right) = x^7 + \frac{1}{x^7} + x + \frac{1}{x}$$

$$\text{Therefore, } x^7 + \frac{1}{x^7} = 623 * (72\sqrt{3}) - 3\sqrt{3} = 44853\sqrt{3}$$

Option A is the correct answer.

14. B

Based on the information provided, we can make the following diagram;



We know that the sum of all entries in the Venn diagram should be 150. Therefore,

$$111 + 75 - 2k + 40 - (k - x) - k + 0 = 150$$

$$76 = 4k - x, \text{ where } x \geq 1$$

We must maximise the number of students who chose Physics but not Mathematics. This number is equal to $75 - 2k$. To maximise this we must minimise k .

$$\text{We have } 4k = 76 + x \text{ or } k = 19 + \frac{x}{4}$$

Since x and k are both integers, x should be divisible by 4. To minimise k , we will take the minimum value of x , which is 4.

$$\text{Therefore, } k = 19 + \frac{4}{4} = 20.$$

Thus, the maximum possible number of students who chose physics but not mathematics, is $75 - 2 * 20 = 35$.

15. D

Since the base of the logarithm has to be greater than zero and cannot be 1, x has to be greater than 3 and cannot be 4. Also, since $x^2 - 9 > 0$, x will be greater than 3.

The equation can be rewritten as;

$$\log_{x-3} (x^2 - 9) - \log_{x-3} (x + 1) = 2$$

Or,

$$\log_{x-3} \frac{x^2 - 9}{x + 1} = 2$$

$$\Rightarrow \frac{(x + 3)(x - 3)}{x + 1} = (x - 3)^2$$

$$\Rightarrow \frac{(x + 3)}{x + 1} = (x - 3)$$

$$\Rightarrow (x + 3) = (x - 3)(x + 1)$$

$$\Rightarrow x + 3 = x^2 - 3x + x - 3$$

$$\Rightarrow x^2 - 3x - 6 = 0$$

The roots of the quadratic above are $\frac{3 \pm \sqrt{3^2 + 24}}{2}$

The negative value of x will not be possible, the positive value that satisfies is $\frac{3 + \sqrt{33}}{2}$

The correct answer is option D.

[▶ VIDEO SOLUTION](#)

16. C

Let the average salary of managers be x and let the average salary of engineers be y . The total salary of all the employees will be $(5 + 25) * 60000 = 1800000$

We have, $5x + 25y = 1800000$... (1)

If the average salary of all the employees increases by 5%, the total salary of all the employees will also increase by 5%, because the total number of employees remains the same. The new total salary will be, $1800000 \times 1.05 = 1890000$

Also, the average salary of all the managers has increased by 20% and has become $1.2x$, we have

$$5 * 1.2x + 25y = 1890000, \text{ or } 6x + 25y = 1890000 \text{ ... (2)}$$

Subtracting equation (1) from equation (2), we get,

$$x = 90000$$

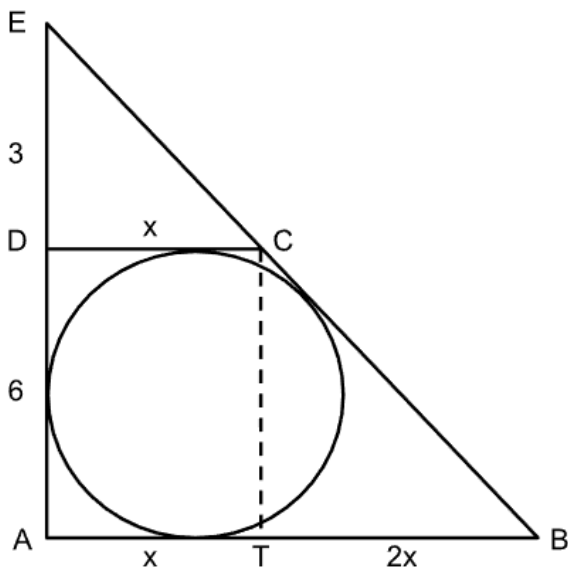
$$\text{Which gives } 25y = 1800000 - 90000 * 5 \text{ or } y = \frac{1350000}{25} = 54000$$

Therefore, the correct answer is option C.

[▶ VIDEO SOLUTION](#)

17. A

We extend BC and AD to meet at E , and based on the information provided, can construct the diagram as follows:



$DC = AT = x$ and $BT = 2x$ such that $AB = 3DC$. $\angle A = 90^\circ$ and triangles ABE and DCE are similar.

Since AD is equal to the diameter of the circle, and $AE = 3DE$, we get $DE = 3$ cm.

In right-angled triangle DCE , we have $CE = \sqrt{x^2 + 9}$

In right-angled triangle BCT , we have $BC = \sqrt{4x^2 + 36} = 2\sqrt{x^2 + 9}$

Since $BE = BC + CE$, we have $BE = 3\sqrt{x^2 + 9}$

The radius of the incircle of a right-angled triangle is given by $\frac{s_1 + s_2 - \text{hypotenuse}}{2}$

Therefore, $3 = \frac{3x + 9 - 3\sqrt{x^2 + 9}}{2}$

Which gives,

$$-x - 1 = -\sqrt{x^2 + 9} \text{ or } x^2 + 2x + 1 = x^2 + 9 \text{ or } x = 4.$$

Lastly, the area of the trapezium would be $\frac{1}{2} \times (x + 3x) \times 6 = 12x = 12 \times 4 = 48$ square centimetres.

[VIDEO SOLUTION](#)

18.112

$$12^{12x} \times 4^{24x+12} \times 5^{2y} = 8^{4z} \times 20^{12x} \times 243^{3x-6}$$

On rewriting after prime factorisation, we get,

$$2^{24x} \times 3^{12x} \times 2^{48x+24} \times 5^{2y} = 2^{12z} \times 2^{24x} \times 5^{12x} \times 3^{15x-30}$$

Since LHS = RHS, the corresponding powers must be equal. We have,

Powers of 3: $12x = 15x - 30$ or $x = 10$, and

Powers of 2: $48x + 24 = 12z$ or $4x + 2 = z$ or $z = 42$

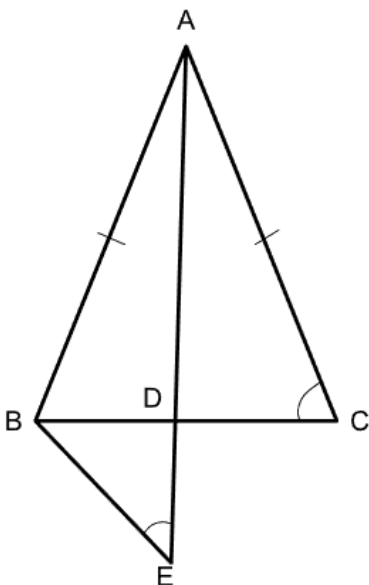
Powers of 5: $2y = 12x$ or $y = 6x$ or $y = 60$

Therefore, $x + y + z = 10 + 42 + 60 = 112$

[VIDEO SOLUTION](#)

19.C

Based on the information provided, we can construct the following diagram;



We have $AB = AC = 12$ cm, and $AD = 8$ cm. And $\angle ACB = \angle AEB$

Since the triangle is isosceles, $\angle ACB = \angle AEB = \angle ABC$ (or $\angle ABD$)

In $\triangle ABE$ and $\triangle ADB$,

$\angle BAE = \angle DAB$ and $\angle AEB = \angle ABD$. Therefore, by AA similarity criterion, $\triangle ABE \sim \triangle ADB$

Hence, we get,

$$\frac{AB}{AD} = \frac{AE}{AB} \text{ or } \frac{12}{8} = \frac{AE}{12} \text{ or } AE = 18 \text{ cm.}$$

 VIDEO SOLUTION

20.16

Vessel A contains 60 litres of pure alcohol, and Vessel B contains 60 litres of pure water. Let the amount taken out from vessel A be x litres. The contents in Vessel A now are $60 - x$ litres of alcohol, and in Vessel B now are 60 litres of water and x litres of alcohol.

After mixing, when the same quantity x is taken out from Vessel B, the alcohol that would be taken out would be $\left(\frac{x}{60+x}\right)^{\text{th}}$ of x , which is $\frac{x^2}{60+x}$ litres.

The initial total quantity of 60 litres has been restored in A after replacement. The total amount of alcohol in Vessel A after the process is complete is:

$$60 - x + \frac{x^2}{60+x} = \frac{(60+x)(60-x) + x^2}{60+x} = \frac{3600}{60+x}$$

The total quantity in vessel A is 60 litres where the ratio of alcohol to total quantity is $\frac{15}{15+4} = \frac{15}{19}$

Therefore,

$$\frac{3600 \div (60+x)}{60} = \frac{15}{19}$$

$$\Rightarrow \frac{60}{60+x} = \frac{15}{19}$$

$$\Rightarrow x = 16$$

Thus, the quantity taken out and replaced in the two iterations is 16 litres.

 VIDEO SOLUTION

21.272

Let the initial total number of coins in box A be $17x$, since the ratio of the number of coins in boxes A and B is $17 : 7$, the initial total number of coins in box B would be $7x$.

After 108 coins are shifted from box A to box B, the number of coins in box A would be $17x - 108$ and in box B would be $7x + 108$. We have,

$$\frac{17x - 108}{7x + 108} = \frac{37}{20}$$

$$\Rightarrow 340x - 2160 = 259x + 3996$$

$$\Rightarrow 81x = 6156 \text{ or } x = 76$$

Therefore, the total number of coins in the two boxes combined are $(17 + 7) * 76 = 1824$, half of which is 912. The number of coins in box B currently is $7 * 76 + 108 = 640$. The number of coins needed to make the coins equal in both the boxes is, $912 - 640 = 272$.

Thus, 272 coins will be transferred from box A to box B to make the number of coins equal (1:1) in both the boxes.

 VIDEO SOLUTION

22. **397**

We have $p + q + r = 900$, and r is a perfect square lying between 150 and 500.

Let q be a non-variable number that does not have a range.

We have $0.3q \leq p \leq 0.7q$, or $q + 0.3q \leq p + q \leq q + 0.7q$ or $1.3q \leq p + q \leq 1.7q$

Also, we can rewrite $p + q = 900 - r$

Therefore, $1.3q \leq 900 - r \leq 1.7q$

Since the extreme values (maximum and minimum) of p depend on the value of q ($0.3q$ and $0.7q$), we can maximise or minimise r to find the extreme values of p .

Since $1.3q \leq 900 - r$, the minimum possible value of q would be when $1.3q = 900 - r$ and when r is maximum. The maximum value of r is 484. We get, $1.3q = 900 - 484$, or $q = 320$.

This gives the minimum possible value of p as $0.3 \times 320 = 96$.

Since $900 - r \leq 1.7q$, the maximum possible value of q would be when $900 - r = 1.7q$ and when r is minimum. The minimum value of r is 169. We get $1.7q = 900 - 169$ or $q = 430$

This gives the maximum possible value of p as $0.7 \times 430 = 301$

The sum of the maximum and minimum values of p is $96 + 301 = 397$.

 VIDEO SOLUTION